

School of Information Technology

Department of Computer Science



COS720 Computer Information & Security I

Project Specification 2026

Release Date: 19 February 2026

Submission Date: 19 May 2026

Lecturer: Mr S.M. Makura

Total: 100 Marks

Instructions

1. This project must be completed on an individual basis. **No group work is allowed.** It must be the **student's own work.** You must clearly indicate if you have objects / components coming from other resources. You must clearly show the uniqueness of your solution in comparison to other existing solutions.
2. An anti-plagiarism approach will be strictly enforced. Please familiarise yourself with the UP-Plagiarism Prevention policy. All plagiarism cases pertaining to any aspect of the project will be handled over to UP Legal for further investigation. You will have to submit a signed form acknowledging that you have read and are in agreement with the **UP Plagiarism Prevention policy which can be read here:**
https://www.up.ac.za/media/shared/1/ZP_Files/s5106-19-plagiarism-prevention-policy.zp181077.pdf).

NO LATE submissions will be accepted after the submission date has lapsed regardless of whatever reason you give.

Project Topic: AI-Powered Insider Threat Detection System

Objective:

Develop a lightweight AI-powered prototype system that detects potential insider threats in a corporate environment by analysing user behavioural features from organisational logs.

The system must classify user activity as:

- Malicious Insider Activity
- Normal / Benign Behaviour

The system must also provide an explanation of why a specific behaviour was flagged as suspicious.

Project Tasks:

1. Research Overview (part of the documentation submitted at the end):

You must include:

- A summary explaining what insider threats are and why they pose a major risk to organisations.
- A brief explanation at most of the page of how Artificial Intelligence and Machine Learning can be applied to behavioural threat detection.
- A comparison between traditional rule-based monitoring and AI-driven detection approaches.

You must also research and summarise the following published paper:

Shoderu, G., Baror, S.O., Makura, S.M. and Modupe, A., 2025. Digital Forensic Readiness to Mitigate Insider Threats in the SaaS Cloud Environment. The Indonesian Journal of Computer Science, 14(6). You can download the paper here:

<http://ijcs.net/ijcs/index.php/ijcs/article/view/5059>

You must:

- Clearly explain **how the paper's findings relate to your solution**
- Identify at least **one gap in the existing literature** that your project attempts to address.

This must be written in **your own words**.

2. Dataset Preparation:

You use the following Kaggle dataset:

Insider Threat Dataset for Behavioral Features & Threat Classification
Dataset for Security Assessment

Available on **Kaggle**

Dataset link: <https://www.kaggle.com/datasets/ahmeduzaki/insider-threat-dataset-for-corporate-environments>

The use of alternative datasets **is not permitted**, in order to ensure fairness and consistency during evaluation.

- Download the prescribed dataset from Kaggle.
- Perform data cleaning and preprocessing.
- Handle missing values and encode categorical variables where applicable.
- Clearly describe:
 - Which features you selected
 - Why those features are relevant to insider threat detection

You must demonstrate understanding of behavioural indicators such as:

- Login time patterns
- File access activity
- Privilege usage
- Abnormal access frequency
- Data transfer behaviour

3. Model Development:

- Implement **at least one** machine learning model (e.g., Logistic Regression, Random Forest, Decision Tree, Support Vector Machine etc).
- Train the model using the provided dataset.
- Evaluate performance using:
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - Confusion Matrix

You must justify:

- Why you selected your model
- Why it is appropriate for insider threat detection

4. **Prototype Implementation:**

Develop a simple prototype system (desktop or web-based) that allows:

- Uploading of a behavioural data record (CSV input), or
- Selection of a sample employee behaviour profile.

The system must output:

- Classification result (Malicious / Normal)
- Confidence score
- A short explanation of key risk indicators contributing to the decision

You are welcome to use **any programming language** you are comfortable with.

5. **AI Explainability:**

Your prototype must include an explainability feature, such as:

- Feature importance ranking
- Highlighting abnormal behavioural indicators
- Risk score breakdown
- Simple human-readable explanation of why the activity was flagged

The explanation must be understandable to a non-technical manager.

6. **Testing Scenarios:**

You must test the system using multiple records from the dataset.

Document:

- True positives
- False positives
- False negatives

Analyse cases where the system performs poorly and suggest realistic improvements.

Submission Requirements:

1. Research and Design Report (Maximum 10 Pages) – to be submitted via the COS720 Project 2026 Turnitin Submission Link

This report must be submitted as a single PDF or Word document and must not exceed **10 pages** (excluding the cover page). It must also **not be zipped**.

The report must include:

1. Generative AI declaration form for assignment submission (Compulsory). Ensure you complete this form.
2. Research Overview
3. Requirements Specification
4. Feature Selection and Dataset Justification
5. Model Selection Rationale
6. UML Design Diagrams:
 - Use Case Diagrams
 - System Architecture Diagram
 - Activity Diagram or Sequence Diagram or Class Diagrams
7. Testing Summary
8. Limitations and Ethical Considerations
9. Bibliography

All UML diagrams **must use** formal UML notation.

2. Technical Documentation (No Page Limit) – to be submitted via the COS720 Project Submission link

This report must be submitted as a separate PDF or Word document and may be zipped depending on the number of files you have. If your zipped file exceeds 100MB (ClickUP maximum file upload size), then upload a document with a Google Drive submission link to where I can view your documentation. Make sure you make the link public so that I can view your files. **Note, this documentation if uploaded to Google Drive after the submission date and time will be excluded from marking.** Below is a summary of what must be in the technical documentation:

1. System Overview
2. Software Requirements and Dependencies
3. Installation Instructions
4. Deployment and Execution Instructions
5. Source Code Documentation
6. Description of Trained Model Files
7. Screenshots Demonstrating Prototype Functionality
8. Troubleshooting Notes (if applicable)

3. Trained AI Model (If Applicable)

Include all trained model files and any preprocessing objects (e.g., encoders, scalers).

4. File Format Requirements

- Documents must be submitted in PDF or Word format (.pdf, .doc, .docx).
- All files which form part of the Technical documentation must be packaged into a single ZIP archive.

Evaluation Criteria:

1. Model Performance

Assessment will focus on:

- Accuracy and effectiveness of insider threat detection
- Precision, Recall, and F1-Score
- Quality of confusion matrix analysis

2. Prototype Functionality

Assessment will focus on:

- Usability of the system
- Correct classification of behavioural records
- Clarity of classification results and confidence scores
- Quality and usefulness of explainability features
- Robustness during demonstration

3. Documentation Quality

Assessment will focus on:

- Clarity and depth of research
- Logical justification of feature and model selection
- Quality of the requirements specification
- Correct and professional UML design
- Quality of testing analysis
- Overall presentation and organisation of the reports

Demonstration Requirement

The project documentation contributes to the final project mark, together with the project demonstration.

Demonstrations will be conducted either:

- Face-to-face with the lecturer, or
- Online.

Failure to demonstrate the project will result in a failing mark for the project (non-negotiable).

Students must ensure that:

- Their prototype runs successfully during the demonstration.
- They understand and can explain all aspects of their implementation.
- Their documentation includes a well-structured UML-based design.

Final Reminder

- **Submission Deadline:** 19 May 2026
- **Late Submissions:** Not accepted
- **Extensions:** Not permitted

Happy coding 😊